

KOLEJ UNIVERSITI TUNKU ABDUL RAHMAN
FACULTY OF COMPUTING AND INFORMATION TECHNOLOGY
ACADEMIC YEAR 2021/2022
DECEMBER EXAMINATION

COMPUTER SCIENCE BACS2163
SOFTWARE ENGINEERING

FRIDAY, 24 DECEMBER 2021

TIME: 9.00 AM – 12.00 NOON (3 HOURS)

BACHELOR OF COMPUTER SCIENCE (HONOURS) IN DATA SCIENCE
BACHELOR OF COMPUTER SCIENCE (HONOURS) IN INTERACTIVE SOFTWARE
TECHNOLOGY
BACHELOR OF COMPUTER SCIENCE (HONOURS) IN SOFTWARE ENGINEERING

BACHELOR OF INFORMATION TECHNOLOGY (HONOURS) IN SOFTWARE SYSTEMS
DEVELOPMENT

Instructions to Candidates:

Answer **ALL** questions in the requested format or template provided.

- This is a final online assessment. You **MUST** answer the assessment questions on your own without any assistance from other persons or resources.
- You must contact the lecturer immediately should there be any disruption to the video conferencing during the final online assessment.
- It is your responsibility to ensure all pages of the answer script are submitted. You will not be notified for any missing pages or incorrect submission of answer script.
- You must submit your answers within the following time frame allowed for this online assessment:
 - The deadline for the submission of your answers is **half an hour** from the end time of this online assessment.
- Penalty **WILL BE IMPOSED** on students who submit their answers late as follows:
 - The final marks of this online assessment will be reduced by 10 marks for answer scripts that are submitted within 30 minutes after the deadline for the submission of answers for this online assessment.
 - The final marks of this online assessment will be downgraded to zero (0) mark for any answer scripts that are submitted after one hour from the end time of this online assessment.
- Extenuating Mitigating Circumstance (EMC) encountered, if any, must be submitted to the Faculty/Branch/Centre within 48 hours after the date of this online assessment. All EMC applications must be supported with valid reasons and evidence. The UC EMC Guidelines apply.

FOCS Additional Instructions to Candidates:

- Include your **FULL NAME, STUDENT ID** and **PROGRAMME OF STUDY** in your submission of answer.
- Read all the questions carefully and understand what you are being asked to answer.
- Marks are awarded for your own (original) analysis. Therefore, use the time and information to build well-constructed answers.

STUDENT'S DECLARATION OF ORIGINALITY

By submitting this online assessment, I declare that this submitted work is free from all forms of plagiarism and for all intents and purposes is my own properly derived work. I understand that I have to bear the consequences if I fail to do so.

BACS2163 SOFTWARE ENGINEERING**Question 1**

Matrix Art Gallery is an international art gallery dedicated to the growth, appreciation, and understanding of modern and contemporary Malaysian art. It strives to appeal to a diverse audience and serve as a resource for both artists and newcomers to the arts. Through exhibitions and public programmes, they hope to bring people closer to art, artists, and their ideas. Matrix Art Gallery has recently made the decision to build a computerised system and has contacted your company as a potential software house to complete this project.

In 80 days, you, as the project manager, must create a computerised system to fulfil all of the user requirements provided by them. The computerised system will allow artists to register as members, pay membership fees, check latest exhibitions, register for programmes, allow the users to chat with company personnel, subscribe to the gallery's newsletter and apply for an internship opportunity. The gallery employees, administrative staff and some in-house artists are all excited about it and will be involved in the development phase of the project. Due to time constraints and improper planning, you will need many experienced software developers to help you with this project.

- a) Determine whether the above-mentioned computerised system is a generic or bespoke software product. Give **TWO (2)** justifications for your answer. (1 + 4 marks)
- b) Recommend and explain **ONE (1)** software process model for developing the above-mentioned computerised system. Explain **FOUR (4)** grounds to back up your recommendation. (2 + 8 marks)
- c) Suggest and explain **THREE (3)** software quality attributes for the above-mentioned computerised system. Justify your recommendations. (6 marks)
- d) As a project manager, you will need to think about how to keep staff motivated throughout the project. Using the *Maslow Motivation Model*, identify and explain **TWO (2)** motivating aspects for this situation. (4 marks)

[Total: 25 marks]

BACS2163 SOFTWARE ENGINEERING**Question 2**

Eternals is a 50-year-old family-owned restaurant located in Kuala Lumpur. The owner of the restaurant finds it challenging to adapt to the new normal of Covid-19 pandemic especially running a small business. Despite having no experience in internet and computing technology, the owner of the restaurant wants to sell their food online. To manage this job, he has approached your organisation, a software house.

- a) Analyse and briefly explain **TWO (2)** ways for determining requirements for the project mentioned above. (4 marks)
- b) Examine **THREE (3)** functional requirements and **TWO (2)** non-functional requirements from the above online ordering system. (5 marks)
- c) Create a Gantt Chart based on the following table to design the online ordering system. Assume the project will begin in February 2022. (9 marks)

Task ID	Task Description	Predecessor	Duration (month)	Overlap (month)
A	Requirement Engineering	None	$\frac{1}{2}$	None
B	System Analysis & Design	A	1	$\frac{1}{4}$
C	Customer Feedback	B	$\frac{1}{2}$	$\frac{1}{4}$
D	Coding	B,C	1	$\frac{1}{4}$
E	Customer Feedback	D	$\frac{1}{2}$	$\frac{1}{4}$
F	Testing	E	$\frac{1}{2}$	None
G	Deployment	F	$\frac{1}{4}$	None

- d) Assume that you are going to use a viewpoint-oriented analysis method to do your requirements analysis. Determine **FIVE (5)** different points of view for the online ordering system. Create a viewpoint hierarchy to present your points of view. (7 marks)

[Total: 25 marks]

BACS2163 SOFTWARE ENGINEERING**Question 3**

Othman Ariff & Associates is a law firm that began its first operation in the 1970s. The firm has a computerised system with a basic text file storage server that keeps a big number of law-related documents. There is no proper documentation for the file storage system, and the modules are not designed properly. The contents of the documentation may be relevant in tracing the chronology of each case in the future. You were recently invited to a meeting with the IT department of Othman Ariff & Associates. The meeting's major purpose is to decide on the legacy systems' maintenance.

- a) Discuss the **TWO (2)** elements that could affect the maintenance cost of the above-mentioned law firm's legacy system based on your expertise. (4 marks)
- b) Using the "Business Value" and "System Quality" categories, explain and justify the technique you should use to keep the law firm's older system running. (3 marks)
- c) Assume that the law firm's IT department has made the decision to re-engineer the legacy system. Propose **ONE (1)** design model for system organization/structuring. Justify and explain your answer. (6 marks)
- d) Propose **SIX (6)** user interface design principles that must be implemented in the legacy system of Othman Ariff & Associates. Justify your recommendations. (12 marks)

[Total: 25 marks]

BACS2163 SOFTWARE ENGINEERING**Question 4**

Kingsman is a well-known smart home developer that offers high-quality home transformations to their clients. Clients may visit Kingsman headquarters to experience the smart home simulation room before transforming their homes. Kingman's Chief Executive Officer plans to computerise a number of services, including smart home simulation room booking, online chatbots and so on. Staff identification cards will be converted to electronic identities (e-IDs) to make it easier for them to access to the smart home simulation room within their office. The smart home simulation room should include the automation of lighting, air-conditioning with infrared remote-control technology, and security system with sensors.

- a) Identify and explain **TWO (2)** sets of stimuli-responses for security system with sensors. (6 marks)
- b) Propose and describe **TWO (2)** functional classifications of Computer Assisted Software Engineering (CASE) tools, each with **TWO (2)** particular examples of tools that can be employed by Kingsman organisation for each classification. (8 marks)
- c) Plan and explain **TWO (2)** testing techniques that can be utilised to test the Kingsman's computerised smart home simulation room booking module. For each testing technique, you must include **TWO (2)** examples of test cases. (11 marks)

[Total: 25 marks]